
GYAN at Full Time

A Four-Expert Ensemble Against the Finished 2026 FIFA World Cup

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The name on the page was Spain.
The name on the trophy is Spain.

Abstract

GYAN (Goal-model, Yield-weighted Aggregation, simulationN) is an ensemble forecasting system built for the 2026 FIFA World Cup, the first 48-team edition of the tournament. It fuses four deliberately heterogeneous experts: a Goal expert built on an Elo-anchored Dixon-Coles goal model with a draw-calibrated correlated negative-binomial score matrix; a Yield expert built from named 26-man squads, individual Transfermarkt values, age weighting, a pro-Europe discount, and a late-injury adjustment; a Socioeconomic expert built on the Hoffmann (2002) specification augmented with FIFA ranking points; and a Market expert built from de-vigged Polymarket, Kalshi, and bookmaker champion prices. The experts are combined in a constrained linear opinion pool fit to historical match-level Ranked Probability Score over the 2014, 2018, and 2022 tournaments, then propagated through 100,000 Monte Carlo simulations of the 48-team bracket. The shipped weights are Goal 5.0%, Yield 5.0%, Socioeconomic 24.4%, and Market 65.6%, with Goal and Yield bound at a structural 5% floor.

The published board placed Spain first at 13.2%, ahead of France, England, and Argentina. Spain won the tournament, unbeaten, 1-0 over Argentina after extra time, their second title after 2010. The four semifinalists were the board's top four, both printed semifinal matchups were exact, and the final standings (Spain, Argentina, England, France) are the board's teams in order 1, 4, 3, 2. Against the strict published checkpoint scoring, the model finished with 33 of 48 exact group positions (68.8%), a Round of 32 field of 27 of 32 (84.4%), knockout winner calls of 24 of 32 (75.0%), and a composite Knockout Score of 42 of 96 (43.8%). This paper gives the methodology its full treatment, restates the evaluation design that was fixed in public before the relevant rounds were played, walks through every checkpoint number against the finished tournament, presents the ablation evidence in full, and closes with the honest interpretive limits of judging a probabilistic forecast against a single realized tournament.

1. Introduction

The pre-tournament technical report opened with a warning that forecasting a World Cup is an exercise in honest humility, and closed with a champion probability of 13.2% for Spain, which is another way of saying the model expected its own headline pick to lose 87 times in 100. Spain won. The temptation after a result like that is to write the victory-lap paper. This is not that paper.

What a finished tournament allows, and what a pre-tournament report cannot have, is a settling of accounts: every published number checked against every played match, with the scoring rules that were fixed in public before the relevant rounds were played. GYAN published its full board, its single most likely bracket, and its methodology on June 10, 2026, before the opening match. It then published four checkpoint reports during the tournament: a Group Stage Review after all 72 group matches, a Knockout Checkpoint after the Round of 32 and Round of 16, a Quarterfinal Checkpoint after the last eight, and a Final Checkpoint after the trophy was lifted. Each one scored the printed page against verified results under rules stated at first publication and never changed afterward, with every figure computed and asserted in code before publication. This paper is the long-form consolidation: the methods in full, the evaluation design, every checkpoint number in its tournament context, the diagnostics that only make sense now that the tournament is over, the ablation evidence that carries the model's independent-value case, and the limitations that survive contact with a good result.

The design of the project was shaped by two pieces of forecasting psychology as much as by any statistical reference. Tetlock (2005) catalogued how confident experts evade falsification after the fact, and Montier (2010) distilled those evasions into a short list of stock excuses. GYAN's answer to both was structural: publish the probabilities, the single path, and the scoring

rules before the tournament, then grade the page in public at fixed checkpoints, whatever the grade turned out to be. The finished tournament is the first time the whole of that design can be read at once.

The one-line honest description of the model, unchanged from before the tournament: the shipped GYAN board is a market-anchored, historically calibrated four-expert pool, with structural expert diversity retained through a positive expert floor and assessed through ablations. It called the champion. It also missed its printed finalist, its third-place call, and roughly half of its exact knockout pairings. Both halves of that sentence are the result.

The paper proceeds as follows. Sections 2 through 5 give the methodology its full treatment: the data registry, the four experts with their governing equations, the constrained opinion pool, and the tournament simulator. Section 6 restates the published forecast. Section 7 documents the evaluation design. Sections 8 through 10 walk through the results stage by stage, from the group tables to the trophy. Section 11 collects the cross-cutting diagnostics, Section 12 presents the ablation evidence in full, and Sections 13 through 15 close with limitations, reproducibility, and conclusions. Appendix A is the complete 32-match knockout ledger; Appendix B is the figure and number registry.

2. Data and provenance

GYAN draws on a registry of cached, hashed raw sources, labeled D1 through D15 in the project conventions. Every raw file is force-pulled and SHA-256 hashed into the run record, so each published number traces to a specific frozen input. Table 1 summarizes the sources that feed the final pipeline.

Table 1. Principal data sources in the final pipeline.

ID	SOURCE	ROLE
D1	martj42 international results	Historical match table and backtests
D2	World Football Elo Ratings (eloratings.net)	Elo formula and reference ratings
D3	FiveThirtyEight SPI archive	External benchmark and validation context
D4	Transfermarkt national-team and player values	Yield expert player values
D5	FIFA/Coca-Cola World Ranking	Socioeconomic augmentation, ranking points
D6	2026 World Cup named squads	Yield expert 26-man squads
D7	Official 2026 schedule and groups	Tournament structure
D8	Hoffmann (2002) coefficients	Socioeconomic prior
D9, D10	Goldman Sachs and PELE published boards	Benchmark comparison
D11	World Bank GDP and population	Socioeconomic features
D12	Country mean annual temperature	Climate term
D13, D14, D15	Polymarket, Kalshi, bookmaker outrights	Market expert

The squad and injury snapshot is frozen for reproducibility at the configured timestamp 2026-06-10T19:00:00Z, with the live market snapshot taken at 2026-06-10T22:32:31Z the same evening. A freeze of this kind is inherently imperfect: it captures teams playing on June 11 and June 12, but teams whose first match falls on June 13 or later may still make permitted squad replacements inside their own 24-hour pre-match windows, which a pre-opening forecast cannot capture. This is treated throughout as a stated limitation, not a reproducibility failure, and it returns in Section 13.

Tournament results used for evaluation come from FIFA's match centre, cross-checked at every checkpoint against at least four independent outlets. Section 7 documents that process, including the stale results that were caught and discarded during the tournament.

3. The four experts

Each expert produces, for every team, a full stage board: probabilities of reaching the Round of 32, the Round of 16, the quarterfinal, the semifinal, the final, and of winning the title. Each expert can also price an individual match as a home/draw/away (W/D/L) probability vector. The four are deliberately heterogeneous: one reads goals, one reads squads, one reads economies, and one reads prices. This section gives each its full specification.

3.1 Goal: the footballing core

Team strength enters first through World Football Elo. A single match updates ratings by

$$R' = R + K G (W - W_e), \text{ with } W_e = 1/(10^{(-d_r/400)} + 1) \text{ and } d_r = (R_h - R_a) + H,$$

where K is a match-importance factor (60 for World Cup finals matches down to 20 for friendlies), G is a goal-difference multiplier, W in $\{1, 0.5, 0\}$ is the result, and H is a 100-point home advantage applied only at genuine home venues. A shootout counts as a draw, and the Elo walk is strictly chronological so that each fixture carries only pre-match information. The initial rating is 1500 and the divisor is 400.

On top of these ratings sits a Dixon-Coles bivariate Poisson goal model. For a fixture the expected goals are

$$\lambda = \exp(\eta * 1[\text{non-neutral}] + \alpha_h - \beta_a), \mu = \exp(\alpha_a - \beta_h),$$

with team attack parameters α , defense parameters β , and a home-field log-goal effect η . The independent-Poisson scoreline matrix is corrected on its four low-score cells by the Dixon-Coles factor τ :

$$\tau(0,0) = 1 - \lambda \mu \rho, \tau(0,1) = 1 + \lambda \rho, \tau(1,0) = 1 + \mu \rho, \tau(1,1) = 1 - \rho.$$

Parameters are fit by maximum likelihood with exponential time decay $\phi(t) = \exp(-\xi t)$ at $\xi = 0.0018$ per day, a weak ridge penalty for small-sample teams, and Elo-derived initial abilities, using L-BFGS-B. The score matrix is truncated at 10 goals per side.

For the tournament simulator the score matrix is replaced by a draw-calibrated correlated negative-binomial form that keeps the same Poisson means but adds a shared Γ tempo shock, giving overdispersion and positive home/away goal correlation:

$$P(x, y) = [\Gamma(k + x + y)/(\Gamma(k) x! y!)] (k/T)^k (\lambda/T)^x (\mu/T)^y, \text{ with } T = k + \lambda + \mu,$$

and shared shape $k = 8$, chosen so the predicted draw rate matches the held-out empirical rate. Any score matrix collapses to a match forecast by summing cells: $P_{\text{home}} \text{ over } x > y, P_{\text{draw}} \text{ over } x = y, P_{\text{away}} \text{ over } x < y.$

The engine choice was validated rather than assumed. On a held-out split of 2,538 recent international matches the three candidate score engines were nearly indistinguishable on RPS (Dixon-Coles 0.16691, plain Poisson 0.16678, calibrated negative-binomial 0.16670). The calibrated matrix was selected because it best matches the observed draw rate (23.84% predicted against 23.72% observed) while preserving the Poisson means. The margin between engines is a reminder worth carrying into every later section: at the match level, most defensible models of international football are close, and the differences that matter live in calibration and structure rather than in engine exotica.

3.2 Yield: named squads, real values, live injuries

The Yield expert is the most methodologically novel component. It begins from the named 26-man 2026 squads and attaches an individual Transfermarkt market value to each player. Player values are adjusted in three ways before aggregation. First, a pro-Europe discount of 30% is applied to players at UEFA clubs, correcting Transfermarkt's well-known European listing premium. Second, an age multiplier (a premium for players 23 and under, neutral through the prime band to 28, then a moderate discount to 32 and a larger discount beyond) tilts toward younger talent at equal value, echoing the PELE treatment. Third, a late-injury layer reads an editable injury snapshot: a named player ruled out contributes nothing, a doubtful player contributes at half value, and every change is recorded in an audit trail. The team feature is then the log of the age-and-injury-adjusted, UEFA-discounted squad value.

Crucially, the Yield expert's value to the ensemble is not claimed through its final weight, which, as Section 4 shows, binds at a structural floor. It is evaluated at the margin through a named-versus-nominal ablation, meaning the difference between the full named-squad treatment and a simpler raw team-value signal, and through team-level movement diagnostics confirming that squad and injury news moves forecasts in sensible directions. That evidence is presented in Section 12; the honest summary is that the margin is small but consistently signed, and that the expert's practical value is transparency about who is actually available to play, not a large shift in the pooled probabilities.

3.3 Socioeconomic: the structural prior

The Socioeconomic expert encodes the idea that footballing strength has economic and demographic roots. It evaluates the published Hoffmann (2002) linear predictor

$$S = \alpha + b_1 g + b_2 g^2 + \eta (c - 14)^2 + k * \text{host} + \phi (\text{Latin} \times \text{population share}),$$

where g is GDP per capita (PPP), c is mean annual temperature with a 14 degree C optimum, host is a host-nation dummy, and the final term is a Latin-football interaction. This prior is standardized and blended with standardized FIFA ranking points in a 0.55/0.45 mix, recovering the Klement-style augmentation that lifts the explained variance well above the bare macro model. The expert changes slowly, knows nothing about form or injuries, and exists to anchor the pool in the structural conditions that produce strong football nations over decades rather than weeks.

3.4 Market: prices, de-vigged and pooled

The Market expert reads three live sources for the 2026 board: Polymarket, Kalshi, and a bookmaker consensus. Each raw vector of implied champion probabilities q is de-vigged proportionally,

$$p_i = q_i / \sum_j q_j,$$

which removes the overround, and the three de-vigged vectors are pooled with fixed source weights (Polymarket 0.40, Kalshi 0.30, bookmaker 0.30). The result is the live blended champion vector used directly in the 2026 board.

The Market expert behaves differently across evaluation surfaces, and this matters for honest interpretation. For the historical backtests there are no raw match prices: cached bookmaker champion outrights (a bookmaker consensus for 2014 and 2018, a William Hill board for 2022) are normalized, converted into log-probability strength ratings, anchored to the pre-tournament Elo mean, and passed through the same rating-to-score model used by the feature experts to produce match W/D/L probabilities. For historical champion scoring the direct outright vector is retained. For 2026, the live champion vector is used directly, while non-champion stage columns are reconstructed by scaling the Stage 2 engine path shape to match it. The Market signal is therefore strongest and cleanest at the champion-probability primitive, and is a market-strength hybrid

at the match level rather than weight on raw match prices. Section 4's discussion of the 65.63% Market weight should be read with that construction in mind.

One source-transfer diagnostic closes the loop between the historical weighting and the live 2026 sources. Because the market expert was weighted historically using bookmaker outrights while the live board blends Polymarket, Kalshi, and bookmaker prices, the final Wednesday run compared the refreshed live blended vector against the older cached market proxy. The check did not trip its 0.08 divergence threshold: the largest champion-probability movement was about 0.00115, with the Netherlands the largest mover. The historical weight transferred across the changed source mix without requiring an additional caveat.

4. Combining the experts: the constrained pool

GYAN combines the four expert boards with a linear opinion pool, a weighted arithmetic mean of the expert probability vectors:

$$p = \sum_e w_e p_e, \text{ with } w_e \geq 0 \text{ and } \sum_e w_e = 1.$$

A log opinion pool (a normalized weighted geometric mean) is computed alongside and would be selected only if it materially improved a held-out calibration check; in the final run the linear pool is selected.

The weights minimize mean match-level Ranked Probability Score over the historical tournaments. For an ordered three-category forecast (home, draw, away) with cumulative forecast F and cumulative outcome O ,

$$RPS = [1/(K - 1)] \sum_{i=1}^{K-1} (F_i - O_i)^2, \text{ with } K = 3.$$

RPS rewards forecasts that put mass near the true ordered outcome and is the standard proper score for football W/D/L prediction (Murphy, 1970). The weight problem is a constrained simplex optimization solved with SLSQP:

$$\text{minimize over } w: (1/N) \sum_{n=1}^N RPS(\sum_e w_e p_{\{e,n\}}, o_n) \text{ subject to } \sum_e w_e = 1 \text{ and } w_e \geq 0.05,$$

over $N = 192$ historical World Cup matches (144 group, 48 knockout) from 2014, 2018, and 2022, with a positive floor of 5% per expert. The historical fit is no-leakage by tournament start date: no information from after a tournament's opening day is used to score that tournament.

The shipped weights: **Goal 5.0%, Yield 5.0%, Socioeconomic 24.37%, Market 65.63%.**

Three framing points, stated before the tournament and unchanged by the result:

1. **The floor is a structural choice, not a data choice.** In the unconstrained reference fit the selected pool sits at approximately Goal 0.0%, Yield 1.1%, Socioeconomic 30.6%, Market 68.2%. Goal and Yield bind at the 5% floor in the shipped fit, so their 5% weights should not be read as the historical RPS objective independently selecting a 5% contribution. The floor keeps all four experts represented for structure and interpretability, and its in-sample RPS cost relative to the unconstrained selected-pool optimum is about 0.000184, which the project considers a fair price for keeping a four-expert structure.
2. **The optimized weights survive out-of-sample rotation.** In-sample mean RPS improves from 0.202667 (equal weight) to 0.195715 (optimized), a relative improvement of about 3.4%, and under the leave-one-tournament-out guard (fit on two tournaments, score the third, rotate) the optimized pool still beats equal weighting, 0.198503 against 0.202667. The optimized weights are therefore not merely an in-sample artifact, though the floor caveat survives in full: Goal and Yield remain floor-bound in the selected fit.

3. **Market dominance is asked for, not discovered.** A pool weighted two thirds to the market should track the market; agreement with the live market is an alignment check, not independent validation. The Market expert receives most of the weight because, on the group-stage-heavy historical objective, market and strength signals simply predicted match outcomes best. That dominance should not be oversold as the model independently discovering a non-market edge; it tracks the market because the calibration asked it to. The independent-value case rests on the ablations, the non-market benchmarks, and the diagnostics of Sections 11 and 12.

5. Tournament simulation

The combined match model is propagated through the real 2026 structure: 12 groups of four, the top two from each group plus the eight best third-placed teams advancing to a Round of 32, then single elimination through five knockout rounds plus a third-place match. Group standings and best-third ranking follow the official tiebreak order, with third-place bracket assignments resolved through the FIFA Annex C combination table, the mechanism that makes the pairing a team receives sensitive to every fine detail of the group tables. Knockout matches are sampled through extra time (goal means scaled to 30/90 of regulation) and, if still level, a near-coin-flip penalty shootout tilted only slightly by the Elo gap and clamped to the 0.40 to 0.60 band.

The final board is the mean over 100,000 independent simulated tournaments, drawn with a parallel-safe seeded generator (global seed 20260611). Convergence is tight: across the top four contenders the champion probability moves by at most 0.17 percentage points between the 50,000-draw and 100,000-draw runs. The simulation also yields aggregate tournament texture that was published in advance and is revisited against the finished tournament in Section 10.4:

Table 2. Stage 2 simulation diagnostics (100,000 draws), published June 10.

Quantity	Value
Knockout upset rate	35.2%
Group-stage draw rate	24.6%
Knockouts reaching extra time	29.0%
Knockouts decided on penalties	16.3%

6. The published forecast

6.1 The champion board

The final pre-opening board, June 10, 2026:

Table 3. Top of the final 2026 champion board.

RANK	TEAM	CHAMPION PROBABILITY
1	Spain	13.23%
2	France	11.20%
3	England	9.19%
4	Argentina	8.67%
5	Portugal	7.69%
6	Brazil	7.67%
7	Germany	4.97%
8	Netherlands	3.74%

The leading group is separated by only a few percentage points, and the top two, Spain at 13.23% and France at 11.20%, are closer to each other than either is to a coin flip on reaching the final. The board's own headline number is an argument for modesty: a 13.2% favorite is a favorite that loses far more often than it wins.

6.2 The printed bracket

For presentation, the model also resolved a single deterministic chalk bracket by ranking every slot on champion probability. That printed path (The Bracket Issue, page 4) sent Spain past France and England past Argentina in the semifinals, Spain over England in the final, and France third over Argentina in a 2022 final rematch. The bracket document stated plainly that this is a presentational device, one snapshot of the distribution, and that real tournaments never run that cleanly. Every checkpoint score in Sections 8 through 10 grades that single printed page, and every checkpoint repeated the same caveat when doing so.

6.3 The published internal disagreement

The pre-tournament report also flagged its single most instructive internal disagreement: the Goal engine alone made Argentina the outright favorite at 20.3%, reflecting Argentina's elite Elo rating and reigning-champion status, while the market-anchored ensemble pulled them to fourth at 8.7% and lifted Spain (about 12% engine-only) to first at 13.2%, with France rising from sixth to second. The bracket document told readers this gap was the most interesting number in the model and worth remembering every time Argentina took the field. Section 11.3 returns to how that aged, because the tournament staged the argument in public and settled it in the strangest available way.

6.4 Texture published in advance

Simulation texture published before the opener, revisited in Section 10.4: knockout upset rate 35.2%, group-stage draw rate 24.6%, knockouts reaching extra time 29.0%, knockouts decided on penalties 16.3%. These four figures were pre-registered in the plain sense: printed before any result existed that could confirm or embarrass them.

7. Evaluation design

Each evaluation rule below was fixed at its first publication and never changed afterward. The group and bracket-formation measures were defined in the Group Stage Review; the 3-point knockout scale was defined in the first knockout checkpoint, after the Round of 32 and Round of 16, and then applied unchanged, and therefore genuinely in advance, to the quarterfinals, the semifinals, the third-place match, and the final.

Group stage. Each group is marked out of 4, one point per team placed in its exact final position (1st, 2nd, 3rd, or 4th), 48 points total. This is a strict test by design: it rewards the precise order, not just naming who advanced. A looser companion measure counts the top-two automatic qualifiers as an unordered set.

Round of 32 bracket formation. A bracket is harder to score than a table, because the right teams can advance and still meet different opponents. It is therefore scored in layers: field accuracy (how many of the 32 printed teams reached the Round of 32), exact-matchup accuracy (how many of the 16 actual ties matched the printed pairs), and the intermediate at-least-one-team and both-teams measures.

Knockout matches. Each of the 32 knockout matches (Round of 32 through the final, including the third-place match) is worth up to 3 points:

- **+1 for the winner call.** The team printed as the winner of that bracket position won the match.
- **+1 for the exact matchup.** Both teams in the tie are the two teams printed, surviving every upstream reshuffle.
- **+1 bonus for the perfect tie.** Both at once: right pairing, right winner. The bonus is deliberate: calling the full tie is the hardest thing a printed bracket can do, so it is worth more than the sum of its parts.

Maximum 96 points. Two edge cases were ruled in the first knockout checkpoint and held for the rest of the tournament. First, a tie containing no printed advancer scores zero and is labeled a no-call rather than a missed pick that never existed; Egypt vs Australia was the only case, and the winner-call percentages are shown both with and without it. Second, when two printed winners collide earlier than the page drew them, the pick for each tie is the winner printed for that bracket slot; Spain vs Portugal in the Round of 16 was the operative case, so Spain (drawn to win that slot) counts as the call there, and Portugal's failed slot is scored where it belongs, in the Switzerland tie that its side of the bracket lost. The third-place match scores against the page like every other tie: the model printed France vs Argentina with France third.

Sources. Every result was taken from FIFA's match centre and cross-checked against at least four of ESPN, Sky Sports, Yahoo Sports, FOX Sports, CBS Sports, Al Jazeera, Olympics.com, UEFA.com, Opta, and CNN, with disagreements resolved in favor of FIFA. The cross-checking earned its keep twice: one stale FOX odds page showing France 3-1 Sweden was discarded when FIFA, UEFA, Sky, Yahoo, and Olympics.com all confirmed 3-0, and several mid-day tables published on June 27 still showed pre-final-match group orders (one had Ghana 2nd and Croatia 3rd in Group L, which the final results reversed) and were discarded in favor of the confirmed final standings.

No re-fitting. The weights, the board, and the printed bracket were frozen at the June 10 artifact. Nothing in this paper uses 2026 outcomes to adjust the model being evaluated, and no weight was re-fit at any point during the tournament.

A note on the composite score. The 3-point layered Knockout Score is a descriptive, journalistic scale built to make a printed bracket auditable in public. It is not a proper scoring rule, and it evaluates the chalk snapshot rather than the probability distribution. It did the job it was designed for, which was to let readers hold the page accountable line by line. Proper-score evaluation of the distribution itself is discussed in Sections 10.5 and 13.

8. Results: the group stage

Published in the Group Stage Review after all 72 group matches, played June 11 to June 27.

8.1 The headline numbers

- **Exact group positions: 33 of 48 (68.8%).** Five perfect groups (B, C, F, I, J), all four slots each.

- **Automatic qualifiers as an unordered set: 19 of 24 (79.2%).** Both qualifiers named correctly in 7 of 12 groups (B, C, F, I, J, K, L).
- **Round of 32 field: 27 of 32 (84.4%).** The five printed teams that missed: South Korea, Czechia, Uruguay, Turkey, Iran. The five arrivals from outside the printed field: South Africa, Australia, Sweden, Cape Verde, Ghana.
- **Exact Round of 32 matchups: 5 of 16 (31.3%)** (Brazil vs Japan, Germany vs Paraguay, Netherlands vs Morocco, England vs DR Congo, USA vs Bosnia). Every actual tie contained at least one printed team (16 of 16); 11 of 16 contained both.

8.2 The full group scorecard

Table 4. Predicted against actual final group tables, all 12 groups.

GROUP	GYAN PREDICTED ORDER (1 TO 4)	ACTUAL FINAL ORDER (1 TO 4)	CORRECT SLOTS	SCORE
A	Mexico, South Korea, Czechia, South Africa	Mexico, South Africa, South Korea, Czechia	1st	1/4
B	Switzerland, Canada, Bosnia, Qatar	Switzerland, Canada, Bosnia, Qatar	all four	4/4
C	Brazil, Morocco, Scotland, Haiti	Brazil, Morocco, Scotland, Haiti	all four	4/4
D	USA, Turkey, Paraguay, Australia	USA, Australia, Paraguay, Turkey	1st, 3rd	2/4
E	Germany, Ecuador, Ivory Coast, Curacao	Germany, Ivory Coast, Ecuador, Curacao	1st, 4th	2/4
F	Netherlands, Japan, Sweden, Tunisia	Netherlands, Japan, Sweden, Tunisia	all four	4/4
G	Belgium, Iran, Egypt, New Zealand	Belgium, Egypt, Iran, New Zealand	1st, 4th	2/4
H	Spain, Uruguay, Cape Verde, Saudi Arabia	Spain, Cape Verde, Uruguay, Saudi Arabia	1st, 4th	2/4
I	France, Norway, Senegal, Iraq	France, Norway, Senegal, Iraq	all four	4/4
J	Argentina, Austria, Algeria, Jordan	Argentina, Austria, Algeria, Jordan	all four	4/4
K	Portugal, Colombia, DR Congo, Uzbekistan	Colombia, Portugal, DR Congo, Uzbekistan	3rd, 4th	2/4
L	England, Croatia, Panama, Ghana	England, Croatia, Ghana, Panama	1st, 2nd	2/4

Total: 33 of 48 (68.8%). Per-group: A 25%, B 100%, C 100%, D 50%, E 50%, F 100%, G 50%, H 50%, I 100%, J 100%, K 50%, L 50%.

8.3 The structure of the misses

The structure of the misses was the finding, and it set the theme for everything that followed.

Most misses were adjacent swaps inside a correctly identified cast. In Groups E, G, H, K, and L the model got the cast of the group right and simply flipped two neighboring places: Ivory Coast and Ecuador swapped 2nd and 3rd in E; Egypt and Iran swapped 2nd and 3rd in G; Cape Verde and Uruguay swapped 2nd and 3rd in H; Colombia edged Portugal for 1st in K; Ghana and Panama swapped 3rd and 4th in L. The model rarely put a team in completely the wrong tier.

Group A was the one genuine scramble. The page had South Korea and Czechia advancing as 2nd and 3rd, with South Africa last. South Africa surged to 2nd and went through, South Korea fell to 3rd and out, and Czechia finished last. Only Mexico at 1st was right, the single weakest group on the card.

A few advancement calls were upset outright. Turkey (predicted 2nd in D) finished last and went home. Uruguay (predicted 2nd in H) finished 3rd and went home. Iran (predicted to advance from G) finished 3rd and went home. In their

place came Australia, Cape Verde, and other sides the model had rated lower.

The five perfect groups are a genuine strength. B, C, F, I, and J came out exactly as drawn, all four slots each. Group J (Argentina, Austria, Algeria, Jordan) and Group I (France, Norway, Senegal, Iraq) were clean top-to-bottom hits.

8.4 The Round of 32 draw the group stage produced

Table 5. The actual Round of 32 draw against the printed pairs.

#	ACTUAL TIE	PRINTED AS THIS EXACT PAIR?
1	South Africa vs Canada	No (printed South Korea vs Canada)
2	Brazil vs Japan	Yes, exact
3	Germany vs Paraguay	Yes, exact
4	Netherlands vs Morocco	Yes, exact
5	Ivory Coast vs Norway	No (printed Ecuador vs Norway)
6	France vs Sweden	No (printed France vs Egypt)
7	Mexico vs Ecuador	No (printed Mexico vs Ivory Coast)
8	England vs DR Congo	Yes, exact
9	Belgium vs Senegal	No (printed Belgium vs Czechia)
10	USA vs Bosnia	Yes, exact
11	Spain vs Austria	No (printed Spain vs Algeria)
12	Portugal vs Croatia	No (printed Colombia vs Croatia and Portugal vs Senegal)
13	Switzerland vs Algeria	No (printed Switzerland vs Austria)
14	Australia vs Egypt	No (printed Turkey vs Iran)
15	Argentina vs Cape Verde	No (printed Argentina vs Uruguay)
16	Colombia vs Ghana	No (printed Colombia vs Croatia)

The 84.4% field accuracy against 31.3% exact-matchup accuracy is the cleanest early statement of the paper’s recurring theme: the model reads team strength well, and the single printed path pays for every small seeding error twice, once in the table and once in the reshuffled draw it causes. The mechanism is structural. The published bracket assumes the favorites finish their groups cleanly, but the Round of 32 pairing a team receives depends on its exact finishing place and on the Annex C third-place assignment table, so even a small upset inside one group reshuffles who meets whom across several others. Three examples drove most of the difference: South Africa finishing 2nd in A instead of South Korea changed Canada’s opponent; Ivory Coast finishing 2nd in E instead of Ecuador scrambled the Mexico, Ecuador, Ivory Coast, Norway corner of the draw (all four teams were correctly in that corner, just paired differently); and Colombia pipping Portugal for 1st in K flipped both of their sides of the bracket.

For readers who want one blended number that mirrors the group score out of 48: one point per correct team in the field (27) plus one bonus point per exact pairing (5) gives 32 of a possible 48 (66.7%).

8.5 The Black Stars clause

The bracket's page 7 made two separate claims about Ghana and was careful to keep them apart. The single most likely bracket put Ghana 4th in Group L and out, with England winning the group and Croatia 2nd in a rematch of their 2018 semifinal. The warmer version, in the page's own words, was that across the full simulation Ghana reach the knockouts far more often than their 0.4% title price suggests, because the new Round of 32 is forgiving to third-placed sides, with a direct caveat that the figure should be read as encouragement, not a ticket.

Ghana finished 3rd in Group L and went through as one of the eight best third-placed teams. The results: Ghana 1, Panama 0; then England 0, Ghana 0; then Croatia 2, Ghana 1. The final Group L table ran England 1st (7 points), Croatia 2nd (6), Ghana 3rd (4 points, level with Croatia on the goal-difference measure but behind on points), Panama 4th and out. England 1st and Croatia 2nd both landed exactly as page 7 said, 2018 rematch framing and all. The standout result was the goalless draw with England, the eventual group winner and a genuine tournament heavyweight; that single point is what carried Ghana into the best-third places. Ghana then met Colombia in the Round of 32 at Arrowhead Stadium in Kansas City on July 3 and lost 1-0, narrowly, as underdogs again.

The scoring read is simple and worth stating plainly because it is the clearest small-scale vindication of the project's design choice to publish the distribution and its hedges rather than the single path alone. The point estimate lost by one place: the printed path said 4th and out, the reality was 3rd and through. The probability framing was vindicated: Ghana advanced through precisely the door page 7 pointed at, the forgiving third-place route, not by winning the group. The cold version of the model lost that round. The warm version, the one named after the striker, won it.

9. Results: the knockout rounds

The knockout stage ran in four published installments: the Round of 32 (June 28 to July 3), the Round of 16 (July 4 to 7), the quarterfinals (July 9 to 11), and the last stage of two semifinals, the third-place match, and the final (July 14 to 19). This section walks each round against the page, then closes the composite.

9.1 The final Knockout Score

Table 6. The complete Knockout Score, all four stages.

STAGE	WINNER CALLS	EXACT MATCHUPS	PERFECT TIES	POINTS
Round of 32	13/16	5/16	3/16	21/48 (43.8%)
Round of 16	5/8	3/8	2/8	10/24 (41.7%)
Quarterfinals	4/4	1/4	1/4	6/12 (50.0%)
Semis, third place, final	2/4	2/4	1/4	5/12 (41.7%)
Total	24/32 (75.0%)	11/32	7/32	42/96 (43.8%)

Where the printed advancer actually took the field (29 of 32 matches), it won 24 of 29 (82.8%). The three matches with no printed advancer on the field: Egypt vs Australia in the Round of 32 (the tournament's only no-call), Morocco vs Canada in the Round of 16 (the printed winner, the Netherlands, never arrived), and Switzerland vs Colombia in the Round of 16 (the printed winner, Portugal, had already been consumed by the Spain collision rule of Section 7, leaving its pathway without a pick on the field).

For the record, the running composite as published at each checkpoint, since every one of these numbers appeared in print during the tournament:

Table 7. The running score, checkpoint by checkpoint.

PUBLISHED AFTER	WINNER CALLS	WHERE ADVANCER WAS IN THE TIE	COMPOSITE
Round of 32 and Round of 16	18/24 (75.0%)	18/21 (85.7%)	31/72 (43.1%)
Quarterfinals	22/28 (78.6%)	22/25 (88.0%)	37/84 (44.0%)
The final	24/32 (75.0%)	24/29 (82.8%)	42/96 (43.8%)

9.2 Round of 32: thirteen calls, two shootouts, one no-call

Table 8. The sixteen Round of 32 matches against the page.

#	RESULT	GYAN'S PRINTED TIE	VERDICT
1	South Africa 0-1 Canada	South Korea vs Canada, Canada to win	different opponent, winner correct
2	Brazil 2-1 Japan	Brazil vs Japan, Brazil to win	EXACT PAIR + WINNER
3	Germany 1-1 Paraguay (Paraguay win 4-3 on pens)	Germany vs Paraguay, Germany to win	exact pair, winner wrong
4	Netherlands 1-1 Morocco (Morocco win 3-2 on pens)	Netherlands vs Morocco, Netherlands to win	exact pair, winner wrong
5	Ivory Coast 1-2 Norway	Ecuador vs Norway, Norway to win	different opponent, winner correct
6	France 3-0 Sweden	France vs Egypt, France to win	different opponent, winner correct
7	Mexico 2-0 Ecuador	Mexico vs Ivory Coast, Mexico to win	different opponent, winner correct
8	England 2-1 DR Congo	England vs DR Congo, England to win	EXACT PAIR + WINNER
9	Belgium 3-2 Senegal (after extra time)	Belgium vs Czechia, Belgium to win	different opponent, winner correct
10	USA 2-0 Bosnia	USA vs Bosnia, USA to win	EXACT PAIR + WINNER
11	Spain 3-0 Austria	Spain vs Algeria, Spain to win	different opponent, winner correct
12	Portugal 2-1 Croatia	Portugal vs Senegal, Portugal to win	different opponent, winner correct
13	Switzerland 2-0 Algeria	Switzerland vs Austria, Switzerland to win	different opponent, winner correct
14	Egypt 1-1 Australia (Egypt win 4-2 on pens)	no tie drawn with either team as winner	no-call
15	Argentina 3-2 Cape Verde (after extra time)	Argentina vs Uruguay, Argentina to win	different opponent, winner correct
16	Colombia 1-0 Ghana	Colombia vs Croatia, Colombia to win	different opponent, winner correct

The scores. Winner calls 13 of 15 (86.7%) where a printed advancer existed in the tie; 13 of 16 (81.3%) counting the no-call as a miss. Exact matchups 5 of 16 (31.3%), as already published in the group stage review. Perfect ties 3 of 16 (18.8%): Brazil over Japan, England over DR Congo, USA over Bosnia. Round of 16 field: 13 of 16 named (81.3%), with Germany, the Netherlands, and Turkey missing and Paraguay, Morocco, and Egypt in their places. Points 21 of 48 (43.8%).

What stood out. Both failed winner calls were coin flips: Germany and the Netherlands both drew 1-1 and both lost from the penalty spot, so neither printed winner lost a match inside 120 minutes. A penalty shootout is close to a pure 50-50, which is about the softest way a forecast can miss, and the model's own simulator prices shootouts inside a 0.40 to 0.60 band for exactly that reason. Second, every team the model was not tracking went out: South Africa, Sweden, Cape Verde, Ghana, and Australia all reached the Round of 32 from outside the printed field, and all five were eliminated in it, with the construction of Australia vs Egypt guaranteeing one outsider-adjacent side a Round of 16 place and Egypt, at least present on the page as a group-stage force, taking it. Third, the opponent-swap pattern arrived: in the 10 ties where the printed winner faced a different opponent than drawn (Canada, Norway, France, Mexico, Belgium, Spain, Portugal, Switzerland, Argentina, Colombia), the printed winner won all 10. The winner calls failed only inside exact pairings, never in reshuffled ones. Section 11.2 tracks that pattern to its eventual break.

9.3 Round of 16: five slots, one true upset

Table 9. The eight Round of 16 matches against the page.

#	RESULT	GYAN'S PRINTED TIE FOR THIS SLOT	VERDICT
1	Morocco 3-0 Canada	Canada vs Netherlands, Netherlands to win	different opponent, slot won, printed team absent
2	France 1-0 Paraguay	Germany vs France, France to win	different opponent, winner correct
3	Norway 2-1 Brazil	Brazil vs Norway, Brazil to win	exact pair, winner wrong
4	England 3-2 Mexico	Mexico vs England, England to win	EXACT PAIR + WINNER
5	Spain 1-0 Portugal	Colombia vs Spain, Spain to win	different opponent, winner correct
6	Belgium 4-1 USA	USA vs Belgium, Belgium to win	EXACT PAIR + WINNER
7	Argentina 3-2 Egypt	Argentina vs Turkey, Argentina to win	different opponent, winner correct
8	Switzerland 0-0 Colombia (Switzerland win 4-3 on pens)	Switzerland vs Portugal, Portugal to win	different opponent, winner wrong

The scores. Winner calls 5 of 8 slots (62.5%): France, England, Spain, Belgium, and Argentina all advanced exactly as printed. The three failed slots: the Netherlands (never arrived; Morocco took the slot and then did what the slot was printed to do, beating Canada), Brazil (lost 2-1 to Norway inside ninety minutes), and Portugal (its pathway fell to Switzerland on penalties). Counting bracket sides rather than team names, 6 of 8 (75.0%). Exact matchups 3 of 8 (37.5%), each drawn a round in advance through two layers of uncertainty: Brazil vs Norway, Mexico vs England, USA vs Belgium. Perfect ties 2 of 8: England over Mexico and Belgium over USA. At least one correct team in the tie, 8 of 8. Quarterfinal field 5 of 8 named (62.5%). Points 10 of 24 (41.7%).

What stood out. The slot pattern earned its name in this round. Morocco vs Canada is the cleanest example: the printed team (the Netherlands) was gone, but the team that removed it stepped into the same bracket position and produced the same printed outcome. And in the same round the pattern's limits showed: Norway beat Brazil in an exactly-predicted tie, and

Switzerland won the tie played in Portugal's pathway. Bracket slots carry real signal because seeding concentrates strength, but a slot is not a guarantee, which is exactly what a probabilistic model would say. Brazil's exit was, and remained to the end, the model's one true knockout upset: Norway winning 2-1 in ninety minutes against a printed semifinal-adjacent pick is the single result that most clearly beat the model's read rather than the bracket's luck.

9.4 Quarterfinals: the round where the page came true

The quarterfinal draw itself arrived with all eight participants somewhere in the printed bracket: not one team reached the last eight from outside the model's knockout picture, and the three the page did not put in the quarterfinals (Morocco, Norway, Switzerland) each occupied the slot of the exact pathway it was drawn losing in.

Table 10. The four quarterfinals against the page.

RESULT	DATE AND VENUE	GYAN'S PRINTED TIE	VERDICT
France 2-0 Morocco	Thu Jul 9, Boston	France vs Netherlands, France to win	different opponent, winner correct
Spain 2-1 Belgium	Fri Jul 10, Los Angeles	Spain vs Belgium, Spain to win	EXACT PAIR + WINNER
England 2-1 Norway (after extra time)	Sat Jul 11, Miami	Brazil vs England, England to win	different opponent, winner correct
Argentina 3-1 Switzerland (after extra time)	Sat Jul 11, Kansas City	Argentina vs Portugal, Argentina to win	different opponent, winner correct

The scores. Winner calls 4 of 4 (100%), the model's first knockout round with zero failed winner calls. Exact matchups 1 of 4 (Spain vs Belgium, the one quarterfinal drawn exactly, called months in advance). Perfect ties 1 of 4 (Spain over Belgium). Points 6 of 12 (50.0%), the best stage percentage of the tournament, and the running composite moved to 37 of 84 (44.0%) with winner calls at 22 of 28 (78.6%), and 22 of 25 (88.0%) where the printed advancer was in the tie.

What stood out. It was not a stroll: England and Argentina both sat level at 1-1 through ninety minutes and won in extra time (scorers, for the record: Mbappe 60 and Dembele 66 for France; F. Ruiz 30 and Merino 88 for Spain with De Ketelaere 41 for Belgium; Bellingham 45+2 and 93 for England with Schjelderup 36 for Norway; Mac Allister 10, Alvarez 112, and Lautaro Martinez 120+1 for Argentina with Ndoye 67 for Switzerland). De Ketelaere's 41st-minute goal was the first Spain conceded all tournament, and it remained the only one. And the round's biggest structural result was the draw it produced: the resulting semifinals, France vs Spain in Dallas and England vs Argentina in Atlanta, matched the printed semifinal draw exactly. Semifinal field 4 of 4, matchups 2 of 2, both called months in advance, with the four semifinalists also the board's top four, so the model's strength read and its bracket read agreed with the tournament at the same moment for the first time.

9.5 Semifinals, third place, and the final: two calls, two misses, one trophy

The printed advancer was on the field in all four ties, and both semifinals were the exact pairings drawn on the page; the third-place match and the final each arrived with one swapped name.

Table 11. The last four matches against the page.

RESULT	DATE AND VENUE	GYAN'S PRINTED TIE	VERDICT
France 0-2 Spain	Tue Jul 14, Dallas	France vs Spain, Spain to win	EXACT PAIR + WINNER
England 1-2 Argentina	Wed Jul 15, Atlanta	England vs Argentina, England to win	exact pair, winner wrong
France 4-6 England (third place)	Sat Jul 18, Miami	France vs Argentina, France third	miss
Spain 1-0 Argentina (after extra time)	Sun Jul 19, New York New Jersey	Spain vs England, Spain champion	winner correct, different opponent

The scores. Winner calls 2 of 4, exact matchups 2 of 4 (both semifinals), perfect ties 1 of 4 (Spain over France). Points 5 of 12 (41.7%). The composite closed at 42 of 96 (43.8%), winner calls at 24 of 32 (75.0%).

The matches, for the record. Spain beat France 2-0 as printed (Oyarzabal 22, a penalty, and Porro 58). Argentina came from behind to beat England 2-1: Gordon 55 put England ahead before E. Fernandez 85 and Lautaro Martinez 90+2 turned it, eliminating the printed finalist at the exact node Section 11.3 discusses. England then beat France 6-4 in a ten-goal third-place match in Miami, England's best World Cup finish since 1966, defeating the printed third-place call: Rice 3, Konsa 18, Saka 37, 45+2, and a late penalty for the hat trick, and Bellingham 90+8 for England; Mbappe 48 and 66, Barcola 54, and Dembele in stoppage time for France, with Mbappe's second making him the World Cup's all-time leading scorer. And Spain beat Argentina 1-0 in extra time at New York New Jersey Stadium to win the title as printed, against an opponent the page did not print: Torres 106, thirty-seven seconds into the second half of extra time, after Argentina's E. Fernandez was sent off in stoppage time. Spain's expected goals read 1.94 from 20 shots to Argentina's 0.20 from 2, and Emiliano Martinez made 11 saves, a World Cup final record. It is Spain's second World Cup title, sixteen years after their first in 2010, and they finished the tournament unbeaten.

9.6 The five losses, decomposed

Across the 29 knockout matches in which the printed advancer took the field, the five losses divide cleanly: two on penalties after 1-1 draws (Germany, the Netherlands), and three inside regulation (Brazil to Norway, England to Argentina, France to England in the third-place match). No printed advancer lost a match that was decided in extra time: every tie settled inside the extra thirty minutes with a printed pick on the field (Belgium and Argentina in the Round of 32, England and Argentina in the quarterfinals, Spain in the final) went to the printed pick. The misses were concentrated in the two places a strength model is most exposed: shootout coin flips, which the model itself prices near even, and single matches between closely rated sides, of which the England vs Argentina semifinal was the archetype the page had flagged in advance.

10. Results: the endgame and the champion

10.1 The four printed endgame calls

Table 12. The endgame against the page.

PRINTED ON PAGE 4	WHAT HAPPENED	SETTLED
Semifinal winners: Spain and England	Spain and Argentina	1 of 2
The final: Spain vs England	Spain vs Argentina	finalists 1 of 2
Third place: France, in a 2022 final rematch	England, best finish since 1966	0 of 1
Champion: Spain, 13.2%	Spain, champions, unbeaten	1 of 1

10.2 The board against the final standings

The tournament's final four were the board's top four, and the final standings map to board ranks as follows: Spain (1st place, board 1), Argentina (2nd, board 4), England (3rd, board 3), France (4th, board 2). Both actual finalists came from the printed top four: Spain first on the champion board at 13.23%, Argentina fourth at 8.67%, either side of France at 11.20% and England at 9.19%. The final the model printed, Spain vs England, was board 1 against board 3; the final the tournament played, Spain vs Argentina, was board 1 against board 4, and it was also the exact final the model's strength-only engine would have leaned toward, with Argentina its outright favorite. Whichever lens is preferred, the trophy match came from the top of the list. The single bracket snapshot scored 43.8% on its composite; the distribution it was drawn from identified the entire top of the tournament.

10.3 Spain's path against the printed path

Spain won Group H as printed (after an opening draw with Cape Verde), beat Austria 3-0 rather than the printed Algeria in the Round of 32, beat Portugal 1-0 rather than the printed Colombia in the Round of 16, then played the script: Belgium in the printed quarterfinal, France in the printed semifinal, and the title, conceding once across all eight matches, De Ketelaere's quarterfinal goal the only breach. Of the model's 32 printed knockout winner calls, the champion accounted for five of the successes, two of them perfect ties (Belgium and France).

10.4 Simulation texture against the actual tournament

Two of the four pre-published texture figures can be checked directly from the 32 verified knockout results. The simulation said 29.0% of knockout matches reach extra time; the actual figure was 9 of 32 (28.1%). The simulation said 16.3% are decided on penalties; the actual figure was 4 of 32 (12.5%). Both realized values sit comfortably inside what a 32-match sample from those rates would produce, and the extra-time figure in particular is close enough to be worth naming as a pre-registered texture call that landed. The group-stage draw rate comparison (24.6% simulated) requires the assembled 72-match group result set and is queued in the production notes; the knockout upset rate depends on a definition of upset that the checkpoints did not fix in advance, so it is left as published rather than graded after the fact.

10.5 A proper-score check on the champion call

The checkpoint composite is descriptive; the champion call can also be scored properly. The realized log loss of a champion forecast is the negative log probability assigned to the actual winner. GYAN assigned Spain 13.23%, a log loss of 2.02 nats. The Goal engine alone, which preferred Argentina and priced Spain near 12%, scores about 2.12 nats. A uniform prior over 48 teams scores 3.87 nats. One event cannot separate models with confidence, and a market-anchored pool beating its own engine on a market-favored outcome is exactly the expected direction; the honest statement is that the shipped board was directionally right at the top and that a single tournament is one draw from the distribution.

11. Diagnostics and patterns

11.1 The board beat the bracket at every stage

The recurring measurement of the tournament, in one table:

Table 13. Field accuracy against exact structure, stage by stage.

STAGE	FIELD/CAST ACCURACY	EXACT STRUCTURE
Group positions	79.2% qualifiers as a set	68.8% exact slots
Round of 32	84.4% field (27/32)	31.3% exact ties
Round of 16	81.3% field (13/16)	37.5% exact ties
Quarterfinals	62.5% field (5/8), all 8 on the printed page	25.0% exact ties
Semifinals	100% field (4/4)	100% exact ties (2/2)
Final	finalists 1 of 2	matchup not exact

The model was consistently strong at naming who belongs at each stage and consistently weaker at the exact pairings, which compound every upstream seeding error. The semifinal row is the exception that proves the mechanism: once the field narrowed to the four teams the model rated strongest, the reshuffling noise ran out and the page and the tournament converged. This was the lesson of every checkpoint, stated at every checkpoint, and it held to the last whistle: the probabilities, not the single printed path, were the model's best product all tournament.

11.2 The swapped-opponent pattern: 17 of 18, and how it broke

Whenever the printed winner faced a different opponent than the one it was drawn against, it won: 10 of 10 in the Round of 32, 3 of 3 in the Round of 16 (France over Paraguay, Spain over Portugal, Argentina over Egypt), 3 of 3 in the quarterfinals (France over Morocco, England over Norway, Argentina over Switzerland), and 1 of 2 in the last stage, closing at 17 of 18 (94.4%). The break was France losing the third-place match to England, at 16 straight; the final restored the pattern one night later when Spain beat the unprinted Argentina.

The caveats published at every checkpoint stand in full. This is a small sample that mostly tracks favorites beating the teams that upset their printed opponents; two of the three quarterfinal cases needed extra time to land; and the pattern broke in exactly the kind of match, a bronze playoff between rotated sides, where favorite status means least. A tendency with a nearly perfect record is still a tendency, not a law, and it should not be promoted into a finding. What it does support, modestly, is the slot logic the checkpoints described: seeding concentrates strength, so a bracket position often carries signal even after the specific name in it has changed.

11.3 The Argentina argument

The pre-tournament report called the engine-versus-ensemble gap on Argentina the single most instructive number in the model: 20.3% engine-only against 8.7% shipped, the largest disagreement between the footballing core and the market-anchored blend anywhere on the board. The tournament then staged the argument in public. The blend's side of the case printed England past Argentina in the semifinal; Argentina won that tie in the 92nd minute, reached the final, and pushed the eventual champion to extra time with ten men. The engine's favorite made the final. The blend's favorite won it, and won the semifinal node where the two lenses disagreed most about the opposite side of the draw.

Neither lens can claim the tournament. The engine, run alone, would have printed Argentina champions and been wrong; the blend printed England into a final England never reached. What can be said, and it is the more useful statement, is that the model identified in advance exactly where its internal disagreement was largest, and that node was where the printed page broke. For a forecasting system, knowing where you are least sure is itself a result, and it is the strongest argument in this tournament for keeping the non-market experts in the pool at more than a floor: not because they moved the shipped probabilities much, but because the disagreement between them and the market was the model's best pre-tournament map of its own fragility.

12. The ablation evidence in full

Nothing in the 2026 outcome changes the historical evidence, which remains the load-bearing case for the four-expert structure. This section collects all of it in one place, because the champion call, however satisfying, is one realized draw, and the ablations are the part of the argument that does not depend on it.

12.1 The headline ablation table

Table 14. Historical mean match-level RPS across 2014, 2018, and 2022 (lower is better).

Pool	MEAN BACKTEST RPS
Shipped GYAN (four experts, constrained weights)	0.195703
Best non-shipped ablation: goal-yield-market (drops Socioeconomic)	0.198709
Equal weights, four experts	0.202667

The shipped four-expert specification beats the best three-expert ablation by about 0.0030 RPS, and it beats equal weighting by about 0.0070. By tournament, the shipped pool's mean match RPS is 0.1849 (2014), 0.1925 (2018), and 0.2096 (2022). The four-expert pool also outperformed a market-only baseline on match RPS in all three backtest tournaments, which is the cleanest available statement that the non-market experts add measurable value at the match level even while the Market expert dominates the weights. The in-sample optimization objective reports the optimized pool at 0.195715 against the same equal-weight 0.202667, a relative improvement of about 3.4%; the Stage 4 backtest recomputes the shipped pool at 0.195703, agreeing to the fourth decimal.

12.2 The out-of-sample guard

Under the leave-one-tournament-out rotation (fit the weights on two tournaments, score the held-out third, rotate through all three), the optimized pool scores 0.198503 against 0.202667 for equal weights. The optimized weights therefore survive the only out-of-sample test the historical record allows. The guard does not erase the floor caveat: Goal and Yield remain floor-bound in the selected fit, and the unconstrained reference optimum (approximately Goal 0.0%, Yield 1.1%, Socioeconomic 30.6%, Market 68.2%) is a permanent reminder that the data, left alone, would have carried even less of the footballing and squad signal than the shipped pool does.

12.3 The Yield margin

The named-versus-nominal ablation isolates what the full Yield treatment (named 26-man squads, individual values, age curve, UEFA discount, injury layer) adds over a simpler raw team-value signal. The Stage 3 named-minus-nominal RPS delta is about 0.000162: small, but consistently signed in the named treatment's favor. The honest reading has not changed since

before the tournament: Yield's contribution is a transparent named-squad and injury adjustment layer whose value shows up most clearly in team-level movement diagnostics when squad and injury news arrives, moving specific forecasts in sensible directions, and whose ensemble weight is the 5% structural floor rather than an earned share. The margin is evaluated at the margin, which is where a component like this lives.

12.4 Expert diversity

The two heavyweight experts are not redundant. The Socioeconomic and Market champion boards correlate at Pearson 0.639 and Spearman 0.697 with a top-12 overlap of 7 of 12, which supports genuine, if partial, structural diversity: both experts can favor rich, strong, high-profile teams, but they do not rank them the same way, and the ablation in Table 14 shows that removing the Socioeconomic expert costs real RPS. The shipped board remains dominated by the combined Socioeconomic-plus-Market share of roughly 90%, which is the honest ceiling on any diversity claim.

12.5 What 2026 adds

These are historical claims, and 2026 adds evidence for the next calibration rather than a re-fit of this one. Folding the 104 matches of 2026 (72 group, 32 knockout) into the historical set would raise the knockout share of a combined future calibration set from 25.0% to 27.0%, a marginal improvement of the objective imbalance discussed in Section 13, and would add the first tournament played under the 48-team structure the simulator was built for. Whether the fitted weights would move, and in which direction, is a question for that future fit, asked with the same no-leakage discipline.

13. Limitations

The limitations published before the tournament survive the good result, and two are added by it.

1. **Market-anchored by design.** Two thirds of the shipped weight listens to the market, so the board's success at the top is substantially the market's success. Agreement between GYAN and the live market benchmark is close to tautological at this weight and is reported throughout as an alignment check. The independent contribution is bounded by the ablation margins of Section 12, not by the champion call.
2. **Objective mismatch.** The weights minimize match-level RPS on a group-stage-heavy historical set (144 of 192 matches), then serve a knockout-sensitive tournament board that lives or dies on bracket paths and variance. This mismatch can partly explain why market and strength signals dominate the fitted weights, and it is a limitation of the available historical calibration target rather than a removable flaw. 2026's 104 matches (72 group, 32 knockout) will raise the knockout share of a combined future calibration set from 25.0% to 27.0%, a marginal improvement of the same imbalance.
3. **The freeze window.** The June 10 squad and injury snapshot (2026-06-10T19:00:00Z) captures teams playing June 11 and June 12 exactly, but could not capture permitted replacements made by later-starting teams inside their own 24-hour pre-match windows. This is inherent to any pre-opening forecast with a reproducible freeze.
4. **Historical market inputs are hybrids.** Backtest market evidence is champion outrights (a bookmaker consensus for 2014 and 2018, a William Hill board for 2022) converted through the rating-to-score model, not raw match prices, and live non-champion stage columns are engine-shape-derived. The 65.63% weight therefore does not mean two thirds weight on observed match markets; the Market signal transfers cleanly only at the champion-probability primitive.
5. **One tournament, one draw.** Every headline result in this paper, including the champion call, is a single realization. A 13.2% call landing is consistent with a well-calibrated board and also with a lucky one; the distinction is not resolvable

from $n = 1$, which is why Sections 10 and 11 lean on rank order, field accuracy, and pre-registered texture rather than the trophy alone.

6. **The composite score is bespoke.** The 3-point knockout scale was designed for public auditability of the printed bracket, and it did that job, but it is not a proper score and rewards only the chalk path. Match-level proper scoring of the 2026 distribution requires per-match model probabilities from the archived run and is queued in the production notes.

14. Reproducibility

The final artifact set is tagged 20260610_223240. The final Stage 3 record (run_stage3_build_ensemble_20260610_223235.json, written 2026-06-10T22:30:40 UTC) and Stage 4 record (run_stage4_evaluation_20260610_223242.json, written 2026-06-10T22:32:40 UTC) capture git commit 0deecdd, global seed 20260611, SHA-256 hashes of every raw input, the configured freeze timestamp 2026-06-10T19:00:00Z with an explicit no-post-freeze-inputs statement, and the live market snapshot time 2026-06-10T22:32:31Z. The pipeline is pure Python, parallelized with joblib, with the hot simulation loops JIT-compiled and a parallel-safe seed sequence so results are stable across worker counts; re-running the staged scripts in order from the recorded commit regenerates the deterministic tables and reproduces the Monte Carlo stages within their documented tolerance.

One provenance gap is recorded honestly: the release tag gyan-v1.0-final does not point at the commit captured in the final run records (the tag points at eee0a29, the records capture 0deecdd, and the repository HEAD at review time was 08e7a2d), so strict tag-level provenance requires either moving the tag to the release commit or creating a new release tag and re-running the staged pipeline from a tagged commit so the records capture it. The input hashes, seed, and commit id in the run records are sufficient to reproduce the shipped artifacts regardless.

All four in-tournament checkpoints computed and asserted their figures in code against verified results before publication: the Group Stage Review, the Round of 32 and Round of 16 Knockout Checkpoint, the Quarterfinal Checkpoint, and the Final Checkpoint. The same discipline applies to this paper: the PDF edition's build script recomputes and asserts every figure in Appendix B's registry, in keeping with the house rule that every published figure is produced and verified programmatically.

15. Conclusion

GYAN set out to be well calibrated, transparent about its sources, and clear about where its information comes from, and to be judged in public against rules it fixed in advance. Judged that way, the finished tournament reads as follows. The board was directionally excellent at the top: the champion, both finalists' provenance, the exact semifinals, and the full final four all came from its first four names. The printed single path performed exactly as a snapshot should be expected to: strong on winner calls (75.0%, and 82.8% where the printed advancer took the field), strict-scored to 43.8% by the pairing layers that punish every upstream reshuffle. The misses were informative rather than random: two shootouts, one genuine upset by Norway, and two defeats at the exact node where the model had flagged its largest internal disagreement. And the honest hedges published in advance, on Ghana, on the swapped-opponent caveats, on the meaning of a 13.2% favorite, all turned out to be the right things to have said.

The name on the page was Spain. The name on the trophy is Spain. The distance between those two sentences and a validated methodology is the entire content of Sections 7 through 12, and it is meant to stay that way.

Disclaimer

This forecast and its evaluation are provided for educational and entertainment purposes only. They are not betting advice, financial advice, or a guarantee of future results. Model outputs are probabilistic estimates based on imperfect data and assumptions, and they should not be used for wagering, investment decisions, or other high-stakes decisions.

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Appendix A: knockout match ledger, all 32 matches

Round of 32. Canada 1-0 South Africa (winner correct, swapped opponent); Brazil 2-1 Japan (perfect tie); Germany 1-1 Paraguay, Paraguay on penalties (exact pair, winner wrong); Netherlands 1-1 Morocco, Morocco on penalties (exact pair, winner wrong); Norway 2-1 Ivory Coast (winner correct, swapped); France 3-0 Sweden (winner correct, swapped); Mexico 2-0 Ecuador (winner correct, swapped); England 2-1 DR Congo (perfect tie); Belgium 3-2 Senegal AET (winner correct, swapped); USA 2-0 Bosnia (perfect tie); Spain 3-0 Austria (winner correct, swapped); Portugal 2-1 Croatia (winner correct, swapped); Switzerland 2-0 Algeria (winner correct, swapped); Egypt 1-1 Australia, Egypt on penalties (no-call); Argentina 3-2 Cape Verde AET (winner correct, swapped); Colombia 1-0 Ghana (winner correct, swapped).

Round of 16. Morocco 3-0 Canada (slot won, printed team absent); France 1-0 Paraguay (winner correct, swapped); Norway 2-1 Brazil (exact pair, winner wrong); England 3-2 Mexico (perfect tie); Spain 1-0 Portugal (winner correct, swapped); Belgium 4-1 USA (perfect tie); Argentina 3-2 Egypt (winner correct, swapped); Switzerland 0-0 Colombia, Switzerland on penalties (winner wrong, Portugal's pathway).

Quarterfinals. France 2-0 Morocco (winner correct, swapped); Spain 2-1 Belgium (perfect tie); England 2-1 Norway AET (winner correct, swapped); Argentina 3-1 Switzerland AET (winner correct, swapped).

Semifinals, third place, final. Spain 2-0 France (perfect tie); Argentina 2-1 England (exact pair, winner wrong); England 6-4 France, third place (miss); Spain 1-0 Argentina AET (winner correct, swapped).

Appendix B: figure and number registry

Every number in this paper traces to one of: the June 10 artifact set (weights, board, backtests, ablations, simulation texture, reproducibility fields), the four published checkpoints (all tournament-facing accuracy figures), or FIFA-verified match results (scores, scorers, extra time, penalties).

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